

Unit 9, Lesson 6: Describing and Interpreting Histograms – Part 1

Benchmark

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

Learning Target

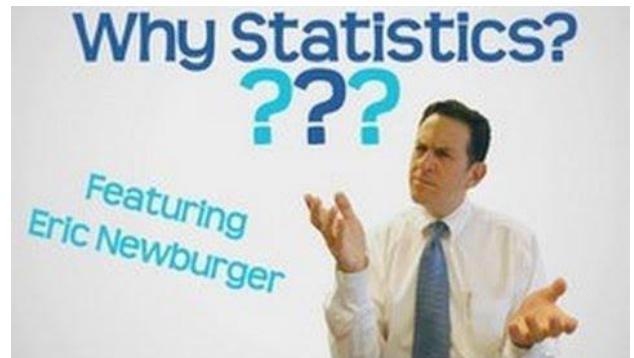
- ☐ I can qualitatively describe and interpret a histogram within a real-world context.

9.6.1 Warm-Up: Why Statistics?

Eric Newburger relates the need for studying statistics to the need for using a microscope. He discusses how a microscope is used to study small things. It's also needed to study things like food poisoning at a larger level to aid in understanding diseases and how viruses spread.

In Mr. Newburger's work, he explains you can take a big thing and shrink it down small enough to study and to understand. That's the study of statistics. And when you want to understand something that is too small, you can use a tool (like the microscope), which is "why statistics."

1. List two reasons why statistics are important.



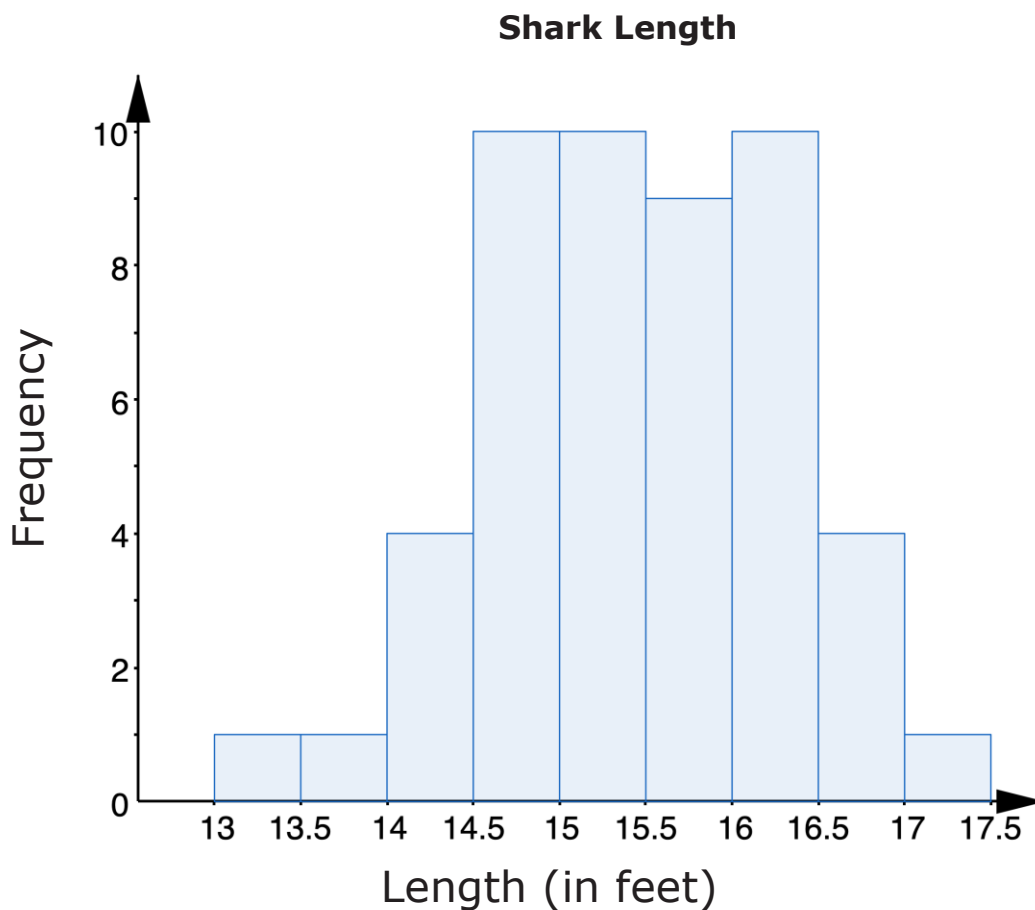
9.6.2 Guided Instruction: Histograms

Numerical data can be displayed graphically in a variety of ways. One type of graph is a histogram.



A **histogram** is a visual display of numerical data using bars along a number line with no spaces between the bars. The height of each bar represents either the frequency or relative frequency of data within that interval.

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



- a. With your partner, discuss the labels for the vertical and horizontal axes.

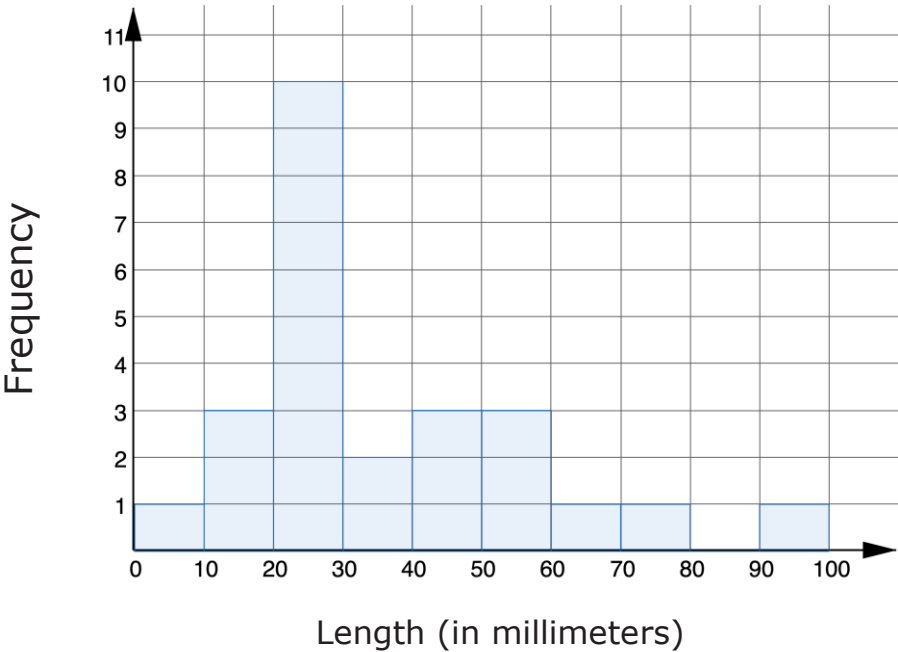
- b. How many sharks were between 15.5 ft and 16 ft?
 - c. How many total sharks were measured to create the histogram?
2. Each bar of a histogram includes the left-end value, but not the right-end value. For example, the first bar includes sharks that are 13 ft and 13.25 ft, but not 13.5 ft.
- a. Discuss with your partner whether or not you can determine the exact lengths of each shark measured from the histogram. Summarize your discussion.
 - b. Discuss with your partner your observations about the width of each bar.
3. If male sharks were added to the sample, hypothesize what would happen to the spread of the data.

9.6.3 Exploration: Histograms

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms’ lengths, in millimeters, in a table and then created a histogram to display the data.

Earthworm Lengths in Millimeters (mm)									
6	11	18	19	20	23	23	25	25	26
27	27	28	29	32	33	41	42	48	52
54	59	60	77	93					

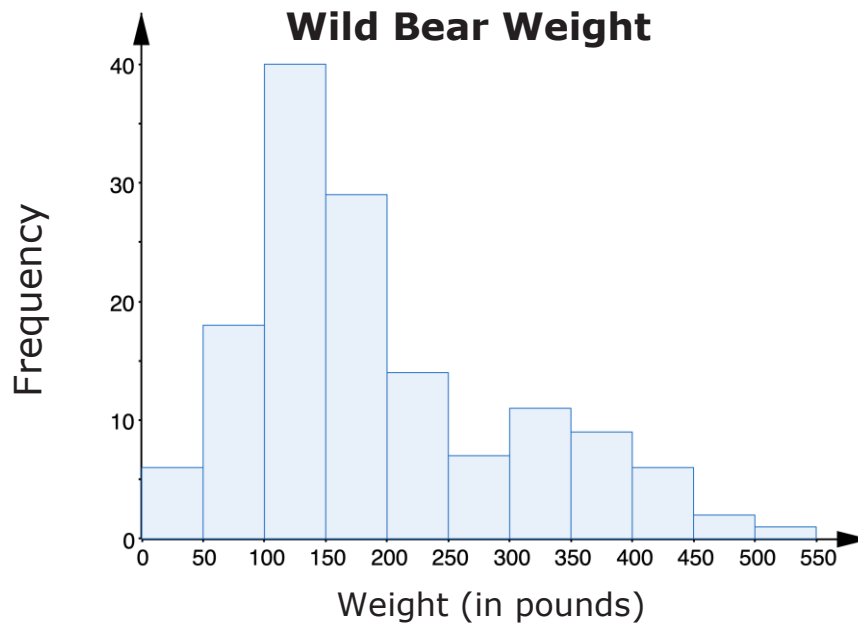


1. How many earthworms were measured in the data?

2. The number of earthworms measured can be determined from
- ☐ the data set.
 - ☐ the histogram.
 - ☐ the data set and the histogram.
3. The length of the shortest earthworm can be determined from
- ☐ the data set.
 - ☐ the histogram.
 - ☐ the data set and the histogram.
4. Place a "*" above the bar that contains the value for the shortest earthworm.
5. Determine the length of the longest earthworm. Then, place a "+" above the bar that contains the value for the longest earthworm.
6. The mean length of the earthworms can be determined from
- ☐ the data set.
 - ☐ the histogram.
 - ☐ the data set and the histogram.
7. What is the mean length of the earthworms?
8. Half of the earthworms are longer than _____ millimeters.
9. Mark an "X" above the bar that contains the mean and mark an "M" above the bar that contains the median.
10. Discuss with your partner how the histogram would change if a 100-millimeter long earthworm was added to the data set. Summarize your discussion.

9.6.4 Try It: Analyzing Histograms

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- Could this sample contain a bear weighing 250 lbs? Explain.
- Mark a "*" above that bar that would contain a 250-lb bear.
- Based on the graph, what can be concluded about the weight of the heaviest bear?
- What is a typical weight for the bears in this graph? Include units.
- Do more than half of the bears in the study weigh less than 250 lbs?

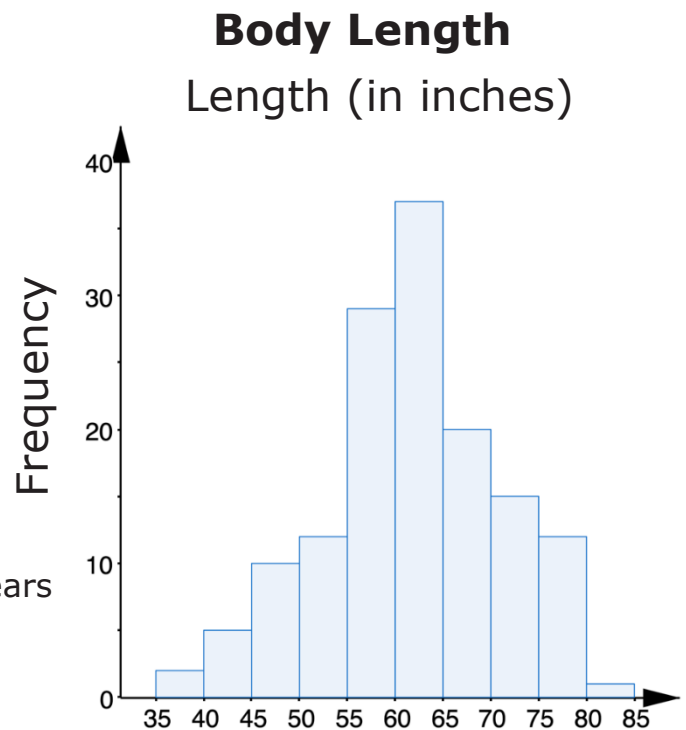
9.6.5 Your Turn!

1. The histogram summarizes the data on the body lengths of the same 143 wild bears.

- a. Does this sample contain a bear that is 85 inches long? Explain your reasoning.

- b. What is a typical length for the bears in this sample? Include units.

- c. Is the spread larger for the weight of bears or the length of bears?



9.6.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate your level of understanding of today's learning target: "I can qualitatively describe and interpret a histogram within a real-world context."

